

Performance Testing

Date	05 July 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Streamlit Testing, Browser Developer Tools
Type of Testing	Functional Testing, Load Testing, Performance Testing
Target Module / API	Voice Upload, AI Analysis Engine, Report Generation
Test Environment	Local System and Streamlit Cloud Deployment
Test Date	06 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Multiple users accessing Streamlit application	5	60	Application loads successfully
2	Uploading and processing voice recordings	5	120	Audio converted and analyzed correctly
3	Generating analysis reports	5	60	Reports generated without errors
4	Viewing dashboard and results	5	60	Charts and feedback displayed properly

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.5 seconds	Pass	Application responded quickly
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	5 req/sec	6 req/sec	Pass	Handled user requests
4	Error Rate	< 1%	0%	Pass	No major failures found
5	CPU Utilization	< 80%	55%	Pass	Stable processing
6	Memory Utilization	< 80%	60%	Pass	Efficient memory usage

Step 4: Observations & Analysis

Key Findings: The VBCUA system successfully processed voice inputs, performed speech-to-text conversion, analyzed concepts, and generated reports. The application showed stable performance with quick response time and accurate output generation.

Bottlenecks Identified: Processing large audio files may take slightly more time due to AI model execution. No major performance issues or application failures were observed during testing.

Optimization Steps Taken: Improved code structure, optimized audio processing workflow, removed unnecessary operations, and maintained modular components for better performance and maintainability.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



